

Bloomberg Businessweek's College Rankings - 2021

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Question of interest

- ▶ **Can we reproduce Bloomberg Businessweek's MBA college ranking?**
- ▶ Dr. Anjani Jain (Yale)

Bloomberg data & methodology - 2021 ranking

- ▶ 119 MBA program (we will look at the first 84)
- ▶ Five variables: Compensation, Learning, Networking, Entrepreneurship, Diversity
- ▶ Weights for the categories are crowd-sourced
- ▶ Surveys: students, alumni, recruiters
- ▶ Key claim: It is intentionally not reproducible!

Data collection and variable creation

- ▶ Survey - each variable: 1-7
- ▶ Raw scores weighted
- ▶ Weighted sum: 1-7
- ▶ Weighted sum is scaled to 0 - 100 ('normalized')
- ▶ Minimum $\rightarrow$ 0, Maximum $\rightarrow$ 100
- ▶ Transformation: $T(x) = \frac{100}{x_{(n)} - x_{(1)}}(x - x_{(1)})$
- ▶ Crowd-sourced weights are

##	Compensation	Learning	Networking	Entre
##	0.357	0.258	0.178	
##	Diversity			
##	0.086			

Toy example - do transformations as Businessweek

```
##  Comp Lear Comp_weighted Lear_weighted w_sum scaled_w_sum
##      2    5           1.2           2.0    3.2         0.0000
##      3    6           1.8           2.4    4.2        83.3333
##      4    5           2.4           2.0    4.4       100.0000

## Compensation      Learning
##           0.6         0.4

## Variance of Compensation is  1
## Variance of Learning is  0.3333333
```

Toy example - do transformations differently

```
##  Comp Lear Comp_scaled Lear_scaled Comp_weighed Lear_we
##      2    5          0          0          0
##      3    6          50         100         30
##      4    5         100          0         60

## Compensation      Learning
##           0.6          0.4

## Variance of Scaled Compensation is  2500
## Variance of Scaled Learning is  3333.333
```

Toy example - do transformations as Businessweek but use z-scores instead

```
##  Comp Lear Comp_z      Lear_z Comp_weigthed Lear_weighted
##      2      5      -1 -0.5773503              -0.6      -0.2309401
##      3      6       0  1.1547005              0.0       0.4618802
##      4      5       1 -0.5773503              0.6      -0.2309401
##  scaled_w_sum
##          0.00000
##        100.00000
##          92.82032

## Compensation      Learning
##           0.6         0.4

## Variance of Variable Z-score is always 1
```

Dataset

##	Rank	School	Overall.score	Compensation
##	1	Stanford	100	100.0
##	2	Dartmouth (Tuck)	92	97.3
##	3	Harvard	89	94.9
##	4	Chicago (Booth)	88	94.5
##	5	Northwestern (Kellogg)	88	94.3
##	5	Columbia	85	93.1

Transformation T is not 1-1

##	x	Tx
----	---	------

##	1	0
----	---	---

##	2	50
----	---	----

##	3	100
----	---	-----

##	x	Tx
----	---	------

##	2	0
----	---	---

##	3	50
----	---	----

##	4	100
----	---	-----

Dr. Jain's main point

Transformation from 1-7 to 0-100 is linear (but he meant affine, not linear).

$$T(x) = \frac{100}{x_{(n)} - x_{(1)}}(x - x_{(1)})$$

$$Z(x) = \frac{x - \mu_x}{\sigma_x}$$

Dr. Jain's claim:

$$Z(T(x)) = Z(x)$$

Let's see if that is correct.

Proof of Dr.Jain's claim

For $a > 0$,

$$Z(ax) = \frac{ax - \mu_{ax}}{\sigma_{ax}} = \frac{ax - a\mu_x}{a\sigma_x} = Z(x)$$

and for any $c \in \mathbb{R}$,

$$Z(x + c) = \frac{x + c - \mu_{x+c}}{\sigma_{x+c}} = \frac{x + c - \mu_x - c}{\sigma_x} = Z(x)$$

Thus,

$$Z(T(x)) = Z\left(\frac{100}{x_{(n)} - x_{(1)}}(x - x_{(1)})\right) = Z(x - x_{(1)}) = Z(x)$$

Dr. Jain's main point is CORRECT.

"If the published 'normalized scores' on each index are simply a 0-100 linear re-scaling of the 1-7 raw data, as BBW's email response indicates, then the published index scores can still be standardized before applying the index weights to them."

We hope to have this:

$$x_i \rightarrow Z(x_i) \rightarrow w_i Z(x_i) \rightarrow \sum_{i=1}^5 w_i Z(x_i) \rightarrow T \left(\sum_{i=1}^5 w_i Z(x_i) \right)$$

$$(1, 7) \rightarrow (-3, 3) \rightarrow (-3w, 3w) \rightarrow (-3, 3) \rightarrow (0, 100)$$

But, we only have scaled data (in the range of 0-100)

Dataset

##	Rank	School	Overall.score	Compensation
##	1	Stanford	100	100.0
##	2	Dartmouth (Tuck)	92	97.3
##	3	Harvard	89	94.9
##	4	Chicago (Booth)	88	94.5
##	5	Northwestern (Kellogg)	88	94.3
##	5	Columbia	85	93.1

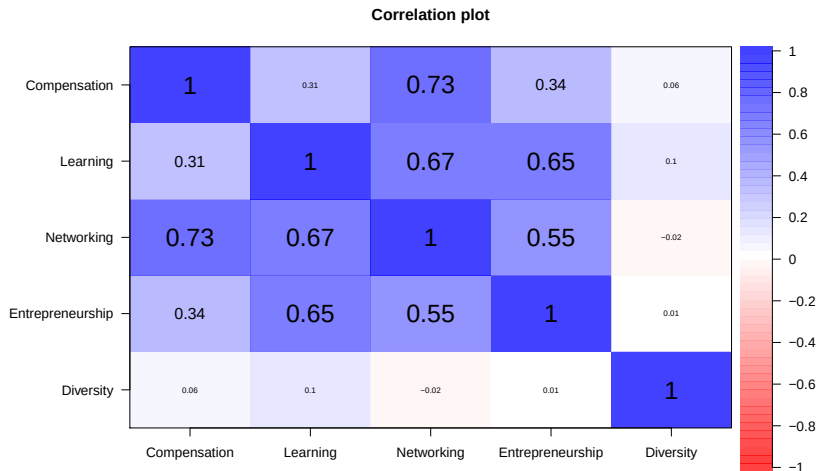
One can still standardize

$$T(x_i) \rightarrow Z(T(x_i)) = Z(x_i)$$

$$Z(x_i) \rightarrow w_i Z(x_i) \rightarrow \sum_{i=1}^5 w_i Z(x_i) \rightarrow T \left(\sum_{i=1}^5 w_i Z(x_i) \right)$$

Step back: understanding data at hand - correlation matrix

MLR will NOT give accurate estimates of true weights



Standardized columns

##	Compensation	Learning	Networking	Entrepreneurship	D
## 1	1.754545	1.3245506	1.563697	2.6702624	0.7
## 2	1.649915	1.0365380	1.269983	0.3435291	0.7
## 3	1.556910	0.3611981	1.036740	0.9375887	0.7
## 4	1.541409	0.5697590	1.174958	1.2346185	0.1
## 5	1.533659	0.5399645	1.399563	0.6405589	0.2
## 6	1.487156	0.4307184	1.235429	0.7890738	0.0

Weighted sum and ranking according to BB weights

##		w_sum	w_sum_normalized	my_rank	BB_rank
##	1	1.635180	100	1	1
##	2	1.189652	87	2	2
##	3	1.011982	82	5	3
##	4	1.064622	84	3	4
##	5	1.032373	83	4	5
##	6	0.959150	81	6	5

Compare rankings

##	my_rank	BB_rank
## 1	1	1
## 2	2	2
## 3	5	3
## 4	3	4
## 5	4	5
## 41	32	39
## 42	54	42
## 43	49	42
## 44	50	44
## 45	47	44
## 80	71	77
## 81	83	81
## 82	70	81
## 83	82	83
## 84	84	84

How many schools have ranking error of more than 3? 5? 10?

```
## # of schools with ranking error of more than three is 56
```

```
## # of schools with ranking error of more than five is 46
```

```
## # of schools with ranking error of more than ten is 26
```

Can we back-engineer the weights? Apply Multiple Linear Regression without intercept

	pr_weights	MLR_weights
## Compensation	0.357	0.73089644
## Learning	0.258	0.10019383
## Networking	0.178	0.02435713
## Entrepreneurship	0.120	0.04766022
## Diversity	0.086	0.06960378

the sum of the coefficients = 0.9727114

R square = 0.9945974

Adjusted R square = 0.9942555

Dr. Jain: “a constrained optimization model that minimizes variances from the published ranking”

Model

$$y = w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + w_5x_5 + \epsilon$$

with $\sum_i w_i = 1$ and $w_i \geq 0$. Then minimize

$$\sum_i (y_i - (w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + w_5x_5))^2$$

Back-engineered weights through constrained optimization

##	pr_weights	const_weights
## Compensation	0.357	0.73752838
## Learning	0.258	0.11863455
## Networking	0.178	0.00000000
## Entrepreneurship	0.120	0.07090125
## Diversity	0.086	0.07293583

LASSO model

```
## R square is 0.9999019
```

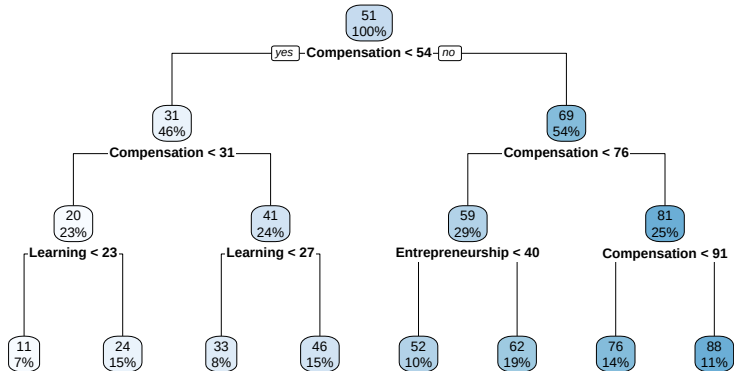
##	lasso_weights	pr_weights
## Compensation	0.67786174	0.357
## Learning	0.06191778	0.258
## Networking	0.10023660	0.178
## Entrepreneurship	0.03064410	0.120
## Diversity	0.06878063	0.086

Compare many models

##	pr_weights	Jain_weights	lasso_weights	
## Compensation	0.357	0.5850	0.67786174	
## Learning	0.258	0.0770	0.06191778	
## Networking	0.178	0.1050	0.10023660	
## Entrepreneurship	0.120	0.0790	0.03064410	
## Diversity	0.086	0.1538	0.06878063	

##	MLR_weights
## Compensation	0.73089644
## Learning	0.10019383
## Networking	0.02435713
## Entrepreneurship	0.04766022
## Diversity	0.06960378

Decision Tree

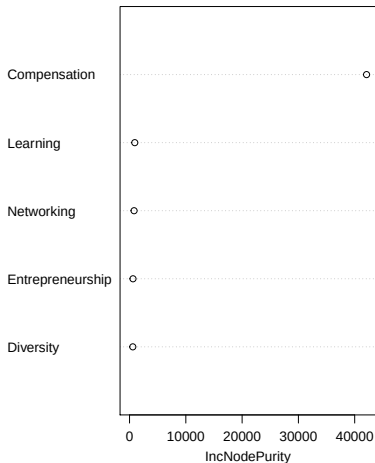


Decision Tree - important variables

##	m1\$variable.importance	perc provi
## Compensation	41847.861	0.40381234
## Networking	26733.308	0.25796395
## Learning	12846.825	0.12396587
## Entrepreneurship	12446.811	0.12010593
## Diversity	9757.146	0.09415191

Bagged (Bootstrap Aggregated) model - important variables

bagged_ranking



Conclusion

Few possibilities

- ▶ BB did not standardize the data (2018, 2019, 2020, 2021)
- ▶ Weights provided were not the true weights used
- ▶ Methodology is not accurately described
- ▶ Provided data is not accurate
- ▶ What else?